of which have adapted recently to severe pollution (see the Perspective by Tobler and Culumber). Sequencing multiple sensitive and resistant populations revealed signals of selective sweeps for variants that may confer tolerance to toxins, some of which were shared between resistant populations. Thus, high genetic diversity in killifish seems to allow selection to act on existing variation, driving rapid adaptation to selective forces such as pollution. —LMZ

Science, this issue p. 1305; see also p. 1232

SYNTHETIC BIOLOGY

Engineering cells to regulate glucose

Diabetes mellitus affects hundreds of millions of people worldwide. Blood glucose levels are chronically deregulated in diabetics, and this can lead to many serious disorders, including cardiovascular disease and renal failure. Xie *et al*. engineered a synthetic circuit into human cells that can sense the glucose concentration and respond to correct deregulation. Implants containing designer cells improved glucose regulation in diabetic mice. —VV

Science, this issue p. 1296

COMPOSITES

Super sensitive, not so silly, putty

Many composites blend stiff materials, such as glass or carbon fibers, into a softer elastic polymer matrix to generate a material with better overall mechanical toughness. Boland *et al*. added graphene to a lightly cross-linked silicone polymer (also known as Silly Putty). The resulting composite has unusual mechanical properties, allowing the manufacture of strain sensors that can detect respiration and the footsteps of spiders. —MSL

Science, this issue p. 1257



Mixing graphene into Silly Putty

PARKINSON'S DISEASE

A mitochondrial target for slowing PD

At present, there are no disease-modifying treatments to stall the progression of Parkinson's disease (PD). A drug in development to treat diabetes might provide a way to slow it down. The drug, MSDC-0160, targets a recently identified carrier of pyruvate (a major substrate for energy production) into mitochondria. Ghosh *et al.* successfully used the drug to block neurodegeneration in several different cellular and animal models of PD. Furthermore, in two mouse models of PD, cellular autophagy was restored and neuroinflammation was reduced. —OMS

Sci. Transl. Med. **8**, 368ra174 (2016).

INFECTIOUS DISEASE

Diseases emerge when forests degrade

Why emerging infectious diseases originate in tropical habitats remains a mystery, but habitat degradation may play a role. *Mycobacterium ulcerans* causes the tropical skin disease Buruli ulcer. Morris *et al.* looked for the presence of *M. ulcerans* in invertebrates and fish that they collected from 17 aquatic sites in French Guiana. They used molecular probes to quantify the mycobacterium and stable isotopes to quantify food web structure. Their findings linked habitat degradation to increases in the abundance of species lower down on the food chain—species more likely to carry the mycobacterium. —SN

Sci. Adv. 10.1126.sciadv.1600387 (2016).

IN OTHER JOURNALS

Edited by **Sacha Vignieri** *and* **Jesse Smith**

METALLURGY

Steel goes for the quick draw

Steel is an important engineering material with properties that change with different processing techniques. Djaziri *et al.* found a new route for strengthening steel by using severe mechanical deformation. The improved strength comes from regions of Fe-C martensite, a known strengthening phase that normally requires quenching from high temperature. Extremely quick drawing of steel wires seems to have a similar effect without requiring high-temperature processing. —BG

Adv. Mater. 10.1002/adma.201601526 (2016).



PHOTO: AMBER/TRINITY COLLEGE DUBLIN; RANDYANDY101/ISTOCKPHOTO

DNA EVENTS

mRNA quality control

DNA is packaged into chromatin, and it must be actively unpacked, or remodeled, for genes to be transcribed into mRNA. That mRNA, synthesized in the nucleus, is also packed onto proteins, forming mRNA ribonucleoparticles (mRNPs). mRNPs are then shunted into the cytoplasm. Babour *et al.* show that one of the chromatin remodeling machines, the enzyme complex ISW1, has a second role in checking that mRNPs are properly packaged and competent to be exported to the cytoplasm. The ISW1 complex binds directly to the premature mRNPs, retaining them in proximity to their transcription sites. —GR

Cell **167**, 1201 (2016).

CANCER IMMUNOTHERAPY

Blocking PI3Kγ makes cold tumors hot

Immunotherapy has changed the face of cancer treatment. However, despite some success, most individuals do not respond to this type of treatment. One potential reason is the presence of immunosuppressive macrophages in the tumor microenvironment. Kaneda *et al*. and De Henau *et al*. now show in mice that inhibiting the lipid kinase phosphoinositide 3-kinase γ (PI3Kγ), or deleting the gene that encodes it, turns immunosuppressive macrophages into immunostimulatory ones in a cell-intrinsic manner. Inhibiting PI3Kγ in tumor-bearing mice led to greater T cell activation within tumors and, when combined with clinically available immunotherapies, to reduced tumor growth and improved overall survival. Clinical trials testing whether this strategy will also work in people are ongoing. —KLM

Nature **539**, 437, 443 (2016).

MICROBIOME

How fiber feeds a healthy gut

Starving bacteria in the gut may turn a beneficial microbial community into one that enhances disease susceptibility. Desai *et al.* explored the effects of eating adequate dietary fiber in a model

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Strong steel is important for infrastructure such as the Golden Gate Bridge.

in which 14 well-characterized strains of human gut bacteria were inoculated into germ-free mice. A diet deficient in fiber increased the abundance of bacteria that digest the polysaccharides in the mucin layer that lines the gut. Thus, instead of helping digest the polysaccharides in dietary fiber, the microbiota attacked the protective layer of the intestinal wall. This allowed an introduced bacterial pathogen to grow and cause lethal colitis. The results may help establish the amounts and types of fiber that best maintain the protective mucin layer in the colon. —LBR

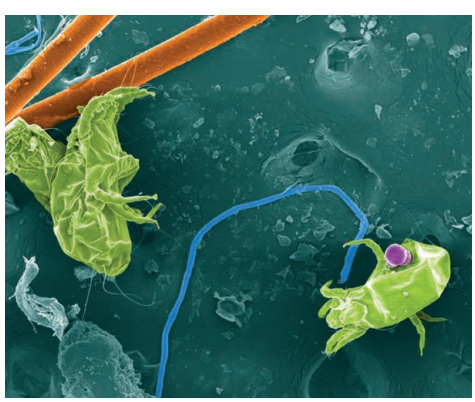
Cell **167**, 1339 (2016).

BIODIVERSITY

Housemates

Whether we recognize it or not, we share our homes with a remarkable diversity of arthropods. Classifying the species that share our space is a challenging task. Madden *et al.* identified over 600 of these species by genotyping house dust collected by citizen scientists from over 700 homes across the United States, revealing the presence of both expected species, such as cockroaches, and unexpected species, such as parasitic wasps. The characteristics of the home or the home occupants were more important for predicting occurrence than were climatic or regional factors, suggesting that home arthropod distributions cannot reliably be predicted from outdoor arthropod predictions but may represent groups of species that are well adapted to human environments. —SNV

Mol. Ecol. 10.1111/mec.13900 (2016).



Genetic screening of dust allows us to identify the arthropods that share our homes.

PHOTO: ANNE A. MADDEN, WITH THE ASSISTANCE FROM ROBERT MCGUGAN/UNIVERSITY OF COLORADO, BOULDER NANOMATERIALS CHARACTERIZATION FACILITY. FALSE-COLORING DONE BY ROBIN HACKER-CARY

METROLOGY

An on-chip cold-atom gravimeter

Gravimeters sense tiny changes in the local gravitational field and are used in geological studies, mineral exploration, and civil engineering to detect underground voids. Typically the size and weight of a small microwave oven, their operation is mechanical: A test mass pulls on a sensitive spring, and any difference in extension is compared with a reference point. Cold atoms cooled to their quantum ground state offer a quantum mechanical approach for detecting gravity. Abend *et al.* have developed an on-chip cold-atom interferometer in which the active part of the sensor, including atom cooling and interference, takes up less than a cubic centimeter. The on-chip approach offers the prospect of compact and sensitive gravimeters for field deployment. —ISO

Phys. Rev. Lett. **117**, 203003 (2016).

MATERIALS CHEMISTRY

Forcing iron to bond to bismuth

Iron-bismuth compounds are of interest for extending the series of superconducting iron-pnictide materials, but the Fe–Bi bond has been elusive. One problem is the immiscibility of Bi in Fe. Walsh *et al*. report that the formation of $FeBi_2$ can be observed under extreme pressure conditions created with diamond anvils. Previous work was unsuccessful with the lower-pressure (body-centered cubic) phase of iron, so pressures sufficient to create its face-centered cubic phase were needed. At pressures above 30 GPa and temperatures of ~1500°C, a phase was observed in x-ray diffraction in which each iron atom coordinated to eight Bi atoms, along with stabilizing Bi-Bi interactions. Once formed, the material was stable down to 2.9 GPa. —PDS

ACS Central Sci. 10.1021/acscentsci.6b00287 (2016).

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An on-chip cold-atom gravimeter

Ian S. Osborne (December 8, 2016)
Science **354** (6317), 1246-1247. [doi: 10.1126/science.354.6317.1246-f]

Editor's Summary

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